

CHAPTER 72

Vitiligo

Rebat M. Halder

Sumayah J. Taliaferro

EPIDEMIOLOGY

Vitiligo occurs worldwide, with a prevalence of 0.1 percent to 2.0 percent. In the United States, the estimated incidence is 1 percent. Vitiligo commonly begins in childhood or young adulthood, with peak onset between 10 and 30 years of age, although it may occur at any age. All races are affected, and both sexes are equally afflicted. A female preponderance has been reported, but this is likely due to increased reporting of cosmetic concerns among female patients.

Familial clustering is common, though inheritance follows a non-Mendelian pattern. Approximately 20 percent of patients have at least one first-degree relative with vitiligo, and the relative risk for first-degree relatives is increased 7- to 10-fold.

ETIOLOGY AND PATHOGENESIS

Vitiligo is a multifactorial polygenic disorder with a complex pathogenesis. Although several theories have been proposed to explain the loss of epidermal melanocytes, the precise cause remains unknown. Significant progress has been made over the past two decades.

Proposed mechanisms include:

- Autoimmune
- Cytotoxic
- Biochemical

- Oxidant-antioxidant imbalance
- Neural
- Viral

These may contribute to the destruction of epidermal melanocytes. Genetic susceptibility also plays a significant role.

Genetics of Vitiligo

Vitiligo is characterized by incomplete penetrance, multiple susceptibility loci, and genetic heterogeneity. Inheritance likely involves genes related to:

- Melanin biosynthesis
- Response to oxidative stress
- Regulation of autoimmunity

The frequent association of vitiligo with autoimmune diseases has led to investigations of HLA associations. HLA types linked to vitiligo in multiple studies include **A2**, **DR4**, **DR7**, and **Cw6**.

Autoimmune Hypothesis and Humoral Immune Response

The association between vitiligo and autoimmune conditions is well established. Commonly associated disorders include:

- **Thyroid disorders:** Hashimoto thyroiditis, Graves disease
- **Endocrinopathies:** Addison disease, type 1 diabetes mellitus
- **Alopecia areata**
- **Pernicious anemia**
- **Systemic lupus erythematosus**
- **Inflammatory bowel disease**
- **Rheumatoid arthritis**
- **Psoriasis**
- **Autoimmune polyglandular syndrome**

The significance of some associations remains debated.

The most compelling evidence for an autoimmune basis is the presence of circulating autoantibodies in vitiligo patients. It remains unclear whether these antibodies are:

- A consequence of melanocyte destruction (epiphenomenon)
- Or active contributors to melanocyte damage

Support for a pathogenic role comes from animal studies showing that pigment cell antibodies can appear before clinical depigmentation.

Smyth Chicken Model

The mutant **Smyth line chicken** is a well-established animal model for studying autoimmune vitiligo. This model has provided insights into aberrant immune processes involved in melanocyte destruction and the development of depigmentation.

VITILIGO AT A GLANCE

- Affects approximately **1 percent of the global population**
- Chronic disorder with **multifactorial predisposition**
- Triggering factors include trauma, sunburn, stress, and systemic illness
- Characterized by **depigmented macules** that may slowly enlarge and new lesions may develop over time
- Most common sites: **periorificial areas, face, genitals, mucous membranes, extensor surfaces, hands, and feet**
- Associated with **autoimmune and endocrine diseases**, and rarely, **malignant melanoma**
- **Fully developed lesions:** absence of melanocytes and inflammatory infiltrate
- **Early lesions:** may show residual melanocytes and mild inflammation